Problem 1: Construct a 2×2 matrix with only one eigenvalue $\lambda = 3$, but which is not diagonalizable. Justify why it fails to be diagonalizable.

Solution

Consider the matrix

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}.$$

The characteristic polynomial is given by

$$\det(A - \lambda I) = \det \begin{pmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{pmatrix} = (3-\lambda)^2.$$

Thus, the only eigenvalue is $\lambda = 3$, and it has algebraic multiplicity 2.
To find the eigenvectors, solve $(A - 3I)\mathbf{x} = 0$, i.e.,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives $x_2 = 0$, and the eigenvector corresponding to $\lambda = 3$ is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
Since there is only one eigenvector for the eigenvalue $\lambda = 3$, the matrix is not diagonalizable. A matrix is diagonalizable if and only if the number of linearly independent eigenvectors equals its size, which is not the case here.

Problem 2: Construct a 2×2 matrix with only one eigenvalue $\lambda = 5$, but which is not diagonalizable. Justify why it fails to be diagonalizable.

Solution

Same as problem 1.

Problem 3: Suppose an $n \times n$ matrix A is invertible, and v is an eigenvector of A with eigenvalue $\lambda \neq 0$. Prove that v is also an eigenvector of A^{-1}. What is the corresponding eigenvalue?

Solution

Given that:

$$Av = \lambda v.$$

Since A is invertible, we can multiply both sides of this equation by A^{-1} to obtain

$$A^{-1}(Av) = A^{-1}(\lambda v).$$

Using the property of inverse matrices, we have

$$v = \lambda A^{-1} v.$$

Thus, we can write

$$A^{-1}v = \frac{1}{\lambda} v.$$

This shows that v is an eigenvector of A^{-1} with eigenvalue $\frac{1}{\lambda}$.

Problem 4: Suppose $A \in \mathbb{R}^{n \times n}$, and v is an eigenvector with eigenvalue λ. Show that v is an eigenvector of A^2, and find the corresponding eigenvalue.

Solution

Let $A \in \mathbb{R}^{n \times n}$, and suppose that v is an eigenvector of A with eigenvalue λ. This means that by definition, we have

$$Av = \lambda v.$$

We need to show that v is an eigenvector of A^2, and find the corresponding eigenvalue.
Consider the action of A^2 on v:

$$A^2 v = A(Av) = A(\lambda v) = \lambda Av.$$

Since $Av = \lambda v$, we substitute this into the equation:

$$A^2 v = \lambda(\lambda v) = \lambda^2 v.$$

Thus, v is indeed an eigenvector of A^2, and the corresponding eigenvalue is λ^2.

Problem 5: Let A be a 2×2 real symmetric matrix. Prove that all eigenvalues of A are real.

Solution

We compute the characteristic equation of $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$:

$$\det\left(\begin{bmatrix} a & b \\ b & c \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \det \begin{bmatrix} a-\lambda & b \\ b & c-\lambda \end{bmatrix} = (a-\lambda)(c-\lambda) - b^2.$$

Expanding this, we get:

$$\lambda^2 - (a+c)\lambda + (ac - b^2) = 0.$$

This is a quadratic equation in λ with the discriminant:

$$\Delta = (a+c)^2 - 4(ac - b^2) = a^2 + 2ac + c^2 - 4ac + 4b^2 = a^2 - 2ac + c^2 + 4b^2.$$

This simplifies to:

$$\Delta = (a-c)^2 + 4b^2.$$

Since both terms $(a-c)^2$ and $4b^2$ are non-negative, we have:

$$\Delta \geq 0.$$

Therefore, the eigenvalues of A are real.